

Practice Quiz: Learning from Every Attempt

Part A: Multiple Choice

1. In Active Inference, a prototype that fails in a specific, unexpected way is: A) A waste of resources that should be avoided B) More informative than a prototype that succeeds as predicted, because it reveals where the generative model is wrong C) Less useful than a successful prototype in all cases D) A sign that the inventor should abandon the project
2. Parameter learning in invention refers to: A) Learning a new programming language B) Adjusting specific values within an existing model structure (e.g., tuning dimensions or concentrations) C) Changing the fundamental causal architecture of the design D) Memorizing technical specifications
3. WD-40 stands for "Water Displacement, 40th formula." This naming reflects: A) Structure learning — the chemist tried 40 fundamentally different approaches B) Parameter learning — the chemist refined the formulation 40 times within the same chemical framework C) Random experimentation without a model D) A marketing decision unrelated to development
4. When persistent prediction errors resist parameter adjustment, the inventor should consider: A) Trying harder with the same approach B) Structure learning — fundamentally changing the model's causal architecture C) Ignoring the errors and shipping the product D) Reducing the number of experiments
5. Netflix's transition from DVD-by-mail to streaming was triggered by: A) A desire to use newer technology for its own sake B) A persistent prediction error — customers wanted immediate access that parameter optimization of mail delivery could not provide C) A government mandate to stop mailing DVDs D) A random management decision
6. A learning protocol for invention includes all of the following EXCEPT: A) Recording predictions before experiments B) Analyzing prediction errors after experiments C) Avoiding documentation to maintain creative spontaneity D) Planning the next experiment based on model updates

7. An inventor is stuck at a local optimum when: A) They have found the globally best solution B) They have found the best solution within their current model structure, but a different structure might produce a better solution C) Their prototype works perfectly under all conditions D) They have run out of ideas entirely

Part B: Short Answer and Analysis

1. An inventor is developing a new plant-based meat substitute. After 20 iterations of adjusting protein ratios, cooking temperatures, and seasoning levels (parameter learning), the product tastes acceptable but the texture remains fundamentally wrong — it is crumbly rather than fibrous. Explain why this texture problem might require structure learning rather than continued parameter learning. What kind of structural change might be needed? (Consider what fundamental causal relationship in the model might be incorrect.)
2. Compare the learning strategies of Thomas Edison (testing thousands of filament materials methodically) and the Wright Brothers (using a wind tunnel to test fundamental aerodynamic assumptions before building full-scale aircraft). Which inventor prioritized parameter learning, and which prioritized structure learning? Under what conditions is each strategy more appropriate? Could either inventor have benefited from adopting the other's strategy earlier?
3. Design a learning protocol for an inventor who is developing a new mobile app for elderly users. The protocol should cover three iteration cycles. For each cycle, specify:
(a) the key uncertainty being tested, (b) whether it is a parameter or structure question, (c) the minimum viable experiment, (d) what outcomes would trigger parameter learning vs. structure learning, and (e) how the learning from this cycle informs the next one.